

AJAY VENKATESH

+1 516 585 9013 Email LinkedIn GitHub Portfolio

Skills

Programming Languages: Python, Java, Scala, R, SQL, C++, TypeScript, JavaScript (ES6)

Big Data & Processing: PySpark, Apache Spark, Pandas, NumPy, ETL/ELT Workflows, Data Pipelines

Databases: SQL Server, MySQL, PostgreSQL, DynamoDB, MongoDB, Dataverse, Oracle, Snowflake

Cloud & Warehouses: AWS (Redshift, S3, EC2, Lambda, Glue, CloudWatch, SageMaker, QuickSight, CDK), GCP (BigQuery), Azure (Data Lake, Function Apps), Azure Data Factory, Azure Data Lake, Kafka

Tools: Git, Docker, Kubernetes, Jupyter, Postman, Power BI, Visual Studio, Cursor, Airflow, Athena, Tableau

AI/ML: TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

Experience

CampusMesh

Jan 2026 – Present

Data Engineer

New York, US

- Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
- Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys, SSM Parameter Store**, and **least-privilege IAM**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Architected a secure data console for Amazon fulfillment center teams to query image/video evidence across FCs; presented **system design** for approval and set up Role Based Access (**4+ roles**) via **OAuth, Midway SSO, HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, **DynamoDB, S3, IAM, CloudWatch**, cutting provisioning time **70%** with **multi-stage, multi-region CI/CD**.
- Engineered distributed backend services in **Java** using Coral, Smithy, Dagger, CBOR; applied **low-level design patterns**, cutting execution time **35%** with unit/integration tests via **Mockito**.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Data Engineer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health platform for **iOS** and **Android**, ingesting real-time wearable telemetry into cloud storage for downstream AI model inference on health metrics and food images.
- Engineered **ETL/ELT workflows** with **Python, PySpark, SQL**, transforming **NoSQL** event data into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
- Built a full-stack insurance analytics platform deployed via **Docker** containers on **EC2**, and implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue across **6 clients**.
- Implemented **document/media processing pipelines** with **WebSocket real-time data streams** to streamline underwriting via a broker portal integrated with insurer platforms.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Data Engineer

Bengaluru, India

- Developed a **distributed data processing backend** on **Microsoft Azure** integrating with **Microsoft CRM**, with **event-driven ingestion** and fault-tolerant retries over **SQL Server** stores.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction from unstructured documents into downstream warehouses, reducing processing time by **75%**.
- Optimized batch ingestion with **parallel processing** for **100K-row** Excel feeds; reduced execution from **5 hours to 1 hour**.
- Built **Power BI** dashboards over datasets joined across multiple relational databases, improving reporting efficiency by **40%**.

Projects

Advanced Expense Tracker (Python Data Pipeline Application) | *Python, SQLite3, Pandas, Matplotlib, Tkinter*

- Architected a modular **Python application** supporting individual and group expense data with secure login, applying **OOP principles** and **MVC & DAO design patterns** over a **SQLite3** persistence layer.
- Implemented concurrent transaction handling and real-time cost aggregation using **Python threading** and locks to ensure thread-safe database writes; visualized spending trends with **Pandas-backed analytics** and **Matplotlib charts**.

Flight Delay Prediction (Big Data) | *Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Matplotlib, Plotly*

- Built a distributed big data pipeline over **Hadoop HDFS** ingesting **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
- Trained **Spark MLlib** classification models to predict delay severity and surfaced feature importance and delay distributions via **Seaborn, Matplotlib, and Plotly** visualizations.

Education

New York University | *Master of Science in Computer Engineering (GPA – 3.9/4)*

Sep 2024 – May 2026

- Coursework:* Big Data, System Design, Machine Learning, Deep Learning, Data Structures

Certifications

AWS Cloud Practitioner

Power BI Data Analyst Associate

Azure Data Fundamentals

Microsoft Developer Associate

Microsoft Solution Architect Expert

Stanford Machine Learning (Coursera)